

The Morphisms of Directed Containers are Cofunctors: completing the dictionary $\mathbf{DCont} \cong \mathbf{Cof}$

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Abstract

The object-level dictionary of `directed-categories.tex` identifies directed containers with small categories. We extend it to morphisms. The container morphism $(f, f^\sharp)$ has its position component $f_s^\sharp: P'(fs) \rightarrow P(s)$ *contravariant* on positions, whereas a functor acts *covariantly* on morphisms; the two variances are opposite, so directed-container morphisms cannot be functors. We resolve the mismatch: with the directed-structure compatibility conditions made fully explicit, $(f, f^\sharp)$ is exactly a *cofunctor* (retrofunctor) of the associated categories. The assignment is a bijection on hom-sets and a functor, and it is bijective on objects, so it is an *isomorphism of categories* $\mathbf{DCont} \cong \mathbf{Cof}$, where $\mathbf{Cof}$ is the category of small categories and cofunctors. The would-be statement $\mathbf{DCont} \simeq \mathbf{Cat}$ is therefore false: the object dictionary does not extend to a full functor into $\mathbf{Cat}$, since $\mathbf{Cof}$ and $\mathbf{Cat}$ have genuinely different hom-sets over their common objects (neither contains the other). As a foil we show that the *opposite* variance — a covariant position map $P(s) \rightarrow P'(fs)$ — recovers functors exactly. All counts are verified combinatorially across 36 ordered pairs of small categories.

1 Setup and the variance problem

We adopt the notation of `directed-categories.tex`. A directed container $D = (S \triangleleft P, \mathbf{o}, \downarrow, \oplus)$ determines a small category $\mathcal{C}(D)$ by the dictionary

$$\mathrm{ob}\mathcal{C}(D) = S, \quad \mathcal{C}(D)(s, s') = \{p \in P(s) : s \downarrow p = s'\}, \quad \mathrm{id}_s = \mathbf{o}(s), \quad q \circ p = p \oplus q,$$

so that $P(s) = \coprod_{s'} \mathcal{C}(D)(s, s')$ is the set of all morphisms *out of* s , and $s \downarrow p = \mathrm{cod}(p)$. Conversely every small category $\mathcal{C}$ arises as $\mathcal{C}(D)$ for a unique directed container D ; this object-level bijection is Theorem 4.1 of `directed-categories.tex`.

A morphism of the underlying containers $(S \triangleleft P) \rightarrow (S' \triangleleft P')$ is a pair

$$(f, f^\sharp), \quad f: S \rightarrow S', \quad f_s^\sharp: P'(fs) \rightarrow P(s) \quad (s \in S),$$

the data of a natural transformation between extension functors. Note the direction: f goes forward on shapes, while $f^\sharp$ pulls positions *backward*, from the target shape fs to the source shape s . Under the dictionary, $f_s^\sharp$ sends a morphism out of fs in $\mathcal{C}(D')$ to a morphism out of s in $\mathcal{C}(D)$.

The problem. A functor $F: \mathcal{C} \rightarrow \mathcal{C}'$ is *covariant* on hom-sets: it sends a morphism out of s to a morphism out of Fs , a map $P(s) \rightarrow P'(Fs)$. The container component $f^\sharp$ runs the other way, $P'(fs) \rightarrow P(s)$. The variances are opposite. Hence the naive hope “a $\mathbf{DCont}$ -morphism is a functor” is ill-typed, and the assertion $\mathbf{DCont} \simeq \mathbf{Cat}$ must be examined rather than assumed. We show the correct target is the category of cofunctors.

2 Cofunctors

Definition 2.1 (Cofunctor). A *cofunctor* $\varphi: \mathcal{C} \leftarrow \mathcal{D}$ between small categories consists of

- an object map $\varphi_0: \text{ob } \mathcal{C} \rightarrow \text{ob } \mathcal{D}$;
- a *lift*: for every object c of $\mathcal{C}$ and every morphism $h: \varphi_0 c \rightarrow d$ of $\mathcal{D}$, a morphism $\varphi^\uparrow(c, h): c \rightarrow c'$ of $\mathcal{C}$;

subject to:

$$\begin{aligned}
(\text{C0}) \quad & \varphi_0(\text{cod } \varphi^\uparrow(c, h)) = \text{cod}(h); & (\text{codomain lift}) \\
(\text{C1}) \quad & \varphi^\uparrow(c, \text{id}_{\varphi_0 c}) = \text{id}_c; & (\text{unit}) \\
(\text{C2}) \quad & \varphi^\uparrow(c, h' \circ h) = \varphi^\uparrow(c', h') \circ \varphi^\uparrow(c, h), \quad c' = \text{cod } \varphi^\uparrow(c, h). & (\text{composition})
\end{aligned}$$

In (C2), $h: \varphi_0 c \rightarrow d$ and $h': d \rightarrow d''$; the term $\varphi^\uparrow(c', h')$ typechecks because $\varphi_0 c' = d = \text{dom}(h')$ by (C0). Cofunctors compose — object maps forward, lifts back — $(\psi\varphi)^\uparrow(c, m) = \varphi^\uparrow(c, \psi^\uparrow(\varphi_0 c, m))$ — with identities $(\text{id}_0, \text{id}^\uparrow)$, forming a category **Cof** of small categories and cofunctors.

Cofunctors are the comonoid morphisms in **(Poly, $\triangleleft$)** (Ahman–Uustalu; Garner); we revisit this in §5.

3 The morphism dictionary

We now write the compatibility conditions that promote a container morphism $(f, f^\#)$ to a morphism of *directed* containers, and read them as the cofunctor laws. For $h \in P'(fs)$, recall $f_s^\#(h) \in P(s)$ and $(fs) \downarrow' h \in S'$.

Definition 3.1 (Directed-container morphism). A *morphism of directed containers* $D \rightarrow D'$ is a container morphism $(f, f^\#)$ satisfying, for all $s \in S$, $h \in P'(fs)$ and $k \in P'((fs) \downarrow' h)$:

$$\begin{aligned}
(\text{M0}) \quad & f(s \downarrow f_s^\#(h)) = (fs) \downarrow' h; & (\downarrow\text{-compatibility}) \\
(\text{M1}) \quad & f_s^\#(\text{o}'(fs)) = \text{o}(s); & (\text{root preservation}) \\
(\text{M2}) \quad & f_s^\#(h \oplus' k) = f_s^\#(h) \oplus f_{s \downarrow f_s^\#(h)}^\#(k). & (\text{shift homomorphism})
\end{aligned}$$

Remark 3.2 (Well-typedness of (M2)). The left side: $h \oplus' k \in P'(fs)$ since $\oplus'$ lands in $P'(fs)$, so $f_s^\#$ applies. The right side: put $p := f_s^\#(h) \in P(s)$. By (M0), $(fs) \downarrow' h = f(s \downarrow p)$, so $k \in P'(f(s \downarrow p))$ and $f_{s \downarrow p}^\#(k) \in P(s \downarrow p)$; then $p \oplus f_{s \downarrow p}^\#(k) \in P(s)$. Thus (M2) typechecks precisely because of (M0) — mirroring how D5 typechecks because of D4.

Lemma 3.3 (Hom-set bijection). *Fix directed containers D, D' with categories $\mathcal{C} = \mathcal{C}(D)$, $\mathcal{C}' = \mathcal{C}(D')$. The map*

$$(f, f^\#) \longmapsto (\varphi_0 := f, \varphi^\uparrow(s, h) := f_s^\#(h))$$

is a bijection between morphisms $D \rightarrow D'$ of directed containers (Definition 3.1) and cofunctors $\mathcal{C} \leftarrow \mathcal{C}'$ (Definition 2.1).

Proof. Under the dictionary, a position $h \in P'(fs)$ is a morphism out of fs in $\mathcal{C}'$, with codomain $(fs) \downarrow' h$; and $f_s^\#(h) \in P(s)$ is a morphism out of s in $\mathcal{C}$, with codomain $s \downarrow f_s^\#(h)$. So the data $(f, f^\#)$ and $(\varphi_0, \varphi^\uparrow)$ are literally the same data. We match the conditions term-by-term.

$(\text{M0}) \Leftrightarrow (\text{C0})$. (M0) reads $f(s \downarrow f_s^\#(h)) = (fs) \downarrow' h$. The left side is $\varphi_0(\text{cod } \varphi^\uparrow(s, h))$ and the right side is $\text{cod}(h)$ in $\mathcal{C}'$. This is exactly (C0).

$(\text{M1}) \Leftrightarrow (\text{C1})$. The identity at fs is $\text{o}'(fs) = \text{id}_{fs}$, and the identity at s is $\text{o}(s) = \text{id}_s$. So $f_s^\#(\text{o}'(fs)) = \text{o}(s)$ is $\varphi^\uparrow(s, \text{id}_{\varphi_0 s}) = \text{id}_s$, exactly (C1).

$(M2) \Leftrightarrow (C2)$. Translate $\oplus$ into $\circ$ by $x \oplus y = y \circ x$. With $p = f_s^\#(h)$ and $s' = s \downarrow p$, the right side of (M2) is $p \oplus f_{s'}^\#(k) = f_{s'}^\#(k) \circ f_s^\#(h) = \varphi^\uparrow(s', k) \circ \varphi^\uparrow(s, h)$, and the left side is $f_s^\#(h \oplus' k) = f_s^\#(k \circ' h) = \varphi^\uparrow(s, k \circ' h)$. So (M2) reads $\varphi^\uparrow(s, k \circ' h) = \varphi^\uparrow(s', k) \circ \varphi^\uparrow(s, h)$ with $s' = \text{cod} \varphi^\uparrow(s, h)$, exactly (C2).

The three conditions are mutually equivalent under an identity map on data, so the assignment is an involution-respecting bijection on hom-sets. $\square$

Lemma 3.4 (Functoriality). *$\mathcal{C}(-)$ together with the assignment of Lemma 3.3 is a functor $\Phi: \mathbf{DCont} \rightarrow \mathbf{Cof}$.*

Proof. Identities. The identity container morphism is (id_S, id) , which maps to $(\text{id}_{\text{ob } \mathcal{C}}, \text{id}^\uparrow)$, the identity cofunctor.

Composition. Let $(f, f^\#): D \rightarrow D'$ and $(g, g^\#): D' \rightarrow D''$. Their container composite has object map $g \circ f$ and position map $(g \circ f)_s^\# = f_s^\# \circ g_{f_s}^\#: P''(gfs) \rightarrow P'(fs) \rightarrow P(s)$ (positions pull back through both stages). Under Φ this is the cofunctor with object map $g_0 \circ f_0$ and lift $m \mapsto f_s^\#(g_{f_s}^\#(m)) = \varphi^\uparrow(s, \psi^\uparrow(\varphi_0 s, m))$, which is exactly the composite cofunctor $\psi\varphi$ of Definition 2.1. Hence Φ preserves composition. $\square$

Theorem 3.5 (Directed containers are categories-and-cofunctors). *The functor $\Phi: \mathbf{DCont} \rightarrow \mathbf{Cof}$ is an isomorphism of categories:*

$$\boxed{\mathbf{DCont} \cong \mathbf{Cof}}$$

Proof. Φ is fully faithful by Lemma 3.3 (bijection on every hom-set) and a functor by Lemma 3.4. On objects, $D \mapsto \mathcal{C}(D)$ is a bijection: by Theorem 4.1 of `directed-categories.tex` every small category is $\mathcal{C}(D)$ for a unique directed container D . A bijective-on-objects, fully faithful functor is an isomorphism of categories (its object bijection and hom bijections assemble into a strict inverse). $\square$

4 Functors live elsewhere: the opposite variance

Theorem 3.5 says the morphisms native to directed containers are cofunctors, not functors. We make the contrast precise and exhibit where functors do sit.

Proposition 4.1 (The dictionary does not extend to **Cat**). *Let $\Phi_0: \text{ob } \mathbf{DCont} \rightarrow \text{ob } \mathbf{Cat}$, $D \mapsto \mathcal{C}(D)$, be the object-level dictionary (a bijection on objects, Theorem 4.1). There is no functor $\mathbf{DCont} \rightarrow \mathbf{Cat}$ that agrees with Φ_0 on objects and is full. In particular the object dictionary does not extend to an equivalence (indeed not to any isomorphism) $\mathbf{DCont} \simeq \mathbf{Cat}$.*

Proof. Let D_1, D_2 be the directed containers of the terminal category **1** and **2** = $\{0 \xrightarrow{u} 1\}$. By Theorem 3.5 (via Lemma 3.3), $|\mathbf{DCont}(D_1, D_2)| = |\mathbf{Cof}(\mathbf{1}, \mathbf{2})|$. We count the latter. A cofunctor $\mathbf{1} \leftarrow \mathbf{2}$ picks $\varphi_0(*) \in \{0, 1\}$ and must lift *every* outgoing morphism of $\varphi_0(*)$ into the unique morphism id_* out of $*$, with (C0) forcing $\varphi_0(\text{cod}(\text{lift})) = \text{cod}$. Object 0 has the outgoing $u: 0 \rightarrow 1$ whose lift would need codomain $\mapsto 1 \neq \varphi_0(*)$: impossible. Object 1 has only id_1 , lifting to id_* . So $|\mathbf{Cof}(\mathbf{1}, \mathbf{2})| = 1$, whence $|\mathbf{DCont}(D_1, D_2)| = 1$. But $|\mathbf{Cat}(\mathbf{1}, \mathbf{2})| = 2$ (choose either object). A functor $G: \mathbf{DCont} \rightarrow \mathbf{Cat}$ with $G = \Phi_0$ on objects would restrict to a map $\mathbf{DCont}(D_1, D_2) \rightarrow \mathbf{Cat}(\mathbf{1}, \mathbf{2})$ from a 1-element set to a 2-element set, which cannot be surjective; so G is not full. As an equivalence (or isomorphism) is in particular full, no such extension exists. $\square$

Remark 4.2 (**Cof** versus **Cat** as abstract categories). Proposition 4.1 is the operative fact for the program: the chain's object equivalence does *not* carry functors. One can ask the sharper question whether **Cof** and **Cat** are equivalent as abstract categories (ignoring the dictionary).

The hom-cardinality data already shows they are not *isomorphic* over their common object class, and moreover the comparison is not even monotone: for the 3-chain $|\mathbf{Cat}(\mathbf{3}, \mathbf{2})| = 4 < 5 = |\mathbf{Cof}(\mathbf{3}, \mathbf{2})|$ while $|\mathbf{Cat}(\mathbf{2}, \mathbf{2})| = 3 > 2 = |\mathbf{Cof}(\mathbf{2}, \mathbf{2})|$ (Table 1), so neither sits inside the other. A full equivalence-level separation requires a relabeling-invariant (e.g. comparing the adjoint profiles of $\text{Hom}(\mathbf{1}, -)$, which is *ob* on **Cat** but the “sink-objects” functor on **Cof**); we do not pursue it here, as Proposition 4.1 already settles the question the dictionary poses.

Proposition 4.3 (Functors = covariant-position morphisms). *Let an op-variance container morphism $D \rightarrow D'$ be (f, ϕ) with $f: S \rightarrow S'$ and a covariant position map $\phi_s: P(s) \rightarrow P'(fs)$ satisfying (N0) $f(s \downarrow p) = (fs) \downarrow' \phi_s(p)$, (N1) $\phi_s(o(s)) = o'(fs)$, and (N2) $\phi_s(p \oplus q) = \phi_s(p) \oplus' \phi_{s \downarrow p}(q)$. Then $(f, \phi) \mapsto F$, $F = f$ on objects, $F(p) = \phi_s(p)$ on a morphism p out of s , is a bijection onto functors $\mathcal{C}(D) \rightarrow \mathcal{C}(D')$.*

Proof. A functor F is precisely: an object map $f = F$; for each s a map of out-sets $\phi_s = F|_{P(s)}: P(s) \rightarrow P'(fs)$; with F preserving codomains (N0: $F(\text{cod}p) = \text{cod}F(p)$), identities (N1: $F(\text{id}_s) = \text{id}_{F(s)}$), and composition (N2: $F(q \circ p) = F(q) \circ F(p)$, i.e. $F(p \oplus q) = \phi_s(p) \oplus' \phi_{s \downarrow p}(q)$). Each datum and law matches under an identity map on data, giving the bijection. $\square$

So functors and cofunctors are the *two opposite variances* of a morphism of directed structure: covariant on positions (push forward, $P(s) \rightarrow P'(fs)$) gives functors; contravariant on positions (pull back, $P'(fs) \rightarrow P(s)$, the genuine container morphism) gives cofunctors. Only the latter is a morphism of containers.

5 The Poly perspective

Under the embedding of directed containers as comonoids in $(\mathbf{Poly}, \triangleleft)$ — the polynomial-comonad incarnation of a small category — container morphisms are exactly comonoid morphisms. Theorem 3.5 is then the statement that comonoid morphisms in $(\mathbf{Poly}, \triangleleft)$ between category-comonoids are cofunctors (Ahman–Uustalu; Garner; Spivak–Garner–Fairbanks). Functors, by Proposition 4.3, are *not* comonoid morphisms; they are a different structure (carried by bicomodules / parametric right adjoints in **Poly**). The variance mismatch flagged at the outset is thus not an accident of bookkeeping but the shadow of two different universal properties: the comonoid maps (cofunctors) and the “functor” maps occupy different hom-objects of **Poly**.

Why this matters for the grant. The equivalence chain advertised in the seed — containers $\simeq$ directed containers $\simeq$ polynomial comonoids $\simeq$ small categories — is an equivalence *of objects*. At the level of morphisms it is an isomorphism $\mathbf{DCont} \cong \mathbf{Cof}$, *not* $\mathbf{DCont} \cong \mathbf{Cat}$. Any application that transports a construction along the chain (supply-chain composition, agent orchestration, the internal replacement theorem) must use cofunctors as the maps. Mistaking them for functors silently changes the morphisms — and, as Proposition 4.1 shows, even their *number*. This is precisely the kind of compositional-correctness pitfall the grant claims to eliminate by formalisation.

6 Computational verification

The Python script `dcont_morphisms_check.py` represents a small category concretely and enumerates, between two categories C, D : (A) functors $C \rightarrow D$; (B) cofunctors $C \leftarrow D$ (Definition 2.1); (C) directed-container morphisms via (M0)–(M2) computed *directly* from the container data $(o, \downarrow, \oplus)$ (Definition 3.1); and (D) op-variance morphisms via (N0)–(N2). Over all 36 ordered pairs from $\{\mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbb{Z}/2, \mathbb{Z}/3, M\}$ (with M the two-element idempotent monoid $\{1, t\}$, $t^2 = t$):

$$\#(C) = \#(B) \text{ in all 36 cases,} \quad \#(D) = \#(A) \text{ in all 36 cases,}$$

confirming Lemma 3.3 and Proposition 4.3; while $\#(A) \neq \#(B)$ in 17 of the 36 cases, confirming Proposition 4.1. A representative excerpt:

$C \rightarrow D$	$\#$ functors	$\#$ cofunctors	$\#$ DCont-morphisms
$\mathbf{1} \rightarrow \mathbf{2}$	2	1	1
$\mathbf{2} \rightarrow \mathbf{2}$	3	2	2
$\mathbf{2} \rightarrow \mathbb{Z}/3$	3	1	1
$\mathbf{3} \rightarrow \mathbf{2}$	4	5	5
$\mathbf{3} \rightarrow \mathbf{3}$	10	6	6
$\mathbb{Z}/3 \rightarrow \mathbb{Z}/3$	3	3	3
$\mathbf{3} \rightarrow M$	4	5	5

Table 1: Functors vs. cofunctors vs. directed-container morphisms. $\#$ cofunctors = $\#$ DCont-morphisms throughout; $\#$ functors differs. $\mathbf{3} \rightarrow \mathbf{2}$ and $\mathbf{2} \rightarrow \mathbf{2}$ show neither of $\mathbf{Cat}$, $\mathbf{Cof}$ dominates the other.

7 Conclusion

The dictionary is now complete at the morphism level. Objects: directed containers = small categories. Morphisms: container morphisms = *cofunctors*. The assignment is an *isomorphism of categories* $\mathbf{DCont} \cong \mathbf{Cof}$. The tempting $\mathbf{DCont} \simeq \mathbf{Cat}$ is false; functors are the opposite variance and are not morphisms of containers. The next step is to formalise $\mathbf{Cof}$ and Theorem 3.5 in Lean (M2/M4), and to revisit every place in the applied notes where “functor between directed containers” was written, replacing it with the correct cofunctor.